THE NOVA RESEARCH CURRICULUM

General Science and Social Studies

Grade 3 · Term 1

Student edition

General Science (4 lessons a week): SC-3.3 solids, liquids and gases; SC-3.2 plants make food; SC-3.7 fair test. Social Studies (3 lessons a week): SS-3.6 directions and maps; SS-3.1 Quaid-e-Azam and Allama Iqbal; SS-3.2 government and citizens. Outside Pakistan, the Social Studies content is replaced with the country's own.

General Science · lesson notes

SC-3.3 Solids, liquids and gases (Weeks 1–4)

1. **Sort:** a stone, water, juice, a pencil, air in a balloon.
2. **Properties:** solids keep their shape; liquids take the shape of the container and flow; gases spread to fill the space.
3. **Change:** ice melts in the hand (solid → liquid); water in the freezer becomes ice (liquid → solid). Safety: only an adult uses heat.

SC-3.2 Plants make their own food (Weeks 5–8)

Plants use sunlight, water and air to make food in their leaves. Leaves are green because of chlorophyll. Show two plants, one in light and one kept in a dark cupboard for a week (starts the fair test).

SC-3.7 Fair test (Weeks 9–12)

Question: *Do plants grow better in sunlight or in the dark?* Change only one thing (light); keep the same pot size, soil, water and plant type. Measure height every 3 days and record it in a table. Draw a bar chart at the end and write one sentence: "I found that ..."

Social Studies · lesson notes

SS-3.6 Directions and maps (Weeks 1–4)

North, south, east and west using a compass rose; the sun rises in the east. Draw a map of the classroom or room at home with a key. Find your province and city on the map of Pakistan.

SS-3.1 Quaid-e-Azam and Allama Iqbal (Weeks 5–8)

- **Quaid-e-Azam Muhammad Ali Jinnah:** born 25 December 1876 in Karachi; led the struggle for Pakistan; first Governor-General; known for "Unity, Faith, Discipline".
- **Allama Muhammad Iqbal:** born 9 November 1877 in Sialkot; national poet; in 1930 he presented the idea of a separate homeland for Muslims of the subcontinent.
- Pakistan came into being on 14 August 1947.

SS-3.2 Government and citizens (Weeks 9–12)

The government makes laws and provides schools, hospitals, roads and safety. Citizens follow laws, keep public places clean, respect others, and vote when they are 18. Make a class rule together and explain why it helps.

▶ **Try it on NovaSchool** (open **Simulations** on your dashboard):

- **Ice, water and steam** — Make it colder or warmer and watch water change between solid, liquid and gas.

Workbook pages

Page	Outcome	Activity
1	SC-3.3	Sort 12 pictures into solid, liquid, gas.
2	SC-3.3	Melting and freezing: draw before and after.
3	SC-3.2	Label what a plant needs to make food.
4–5	SC-3.7	Fair-test record sheet and bar chart.
6	SS-3.6	Compass rose; draw a map of your room with a key.
7	SS-3.6	Colour the provinces of Pakistan; mark your city.
8	SS-3.1	Fact file: Quaid-e-Azam and Allama Iqbal.
9	SS-3.2	Who helps us? Match government services to pictures.

Question bank

General Science

#	Outcome	Question	Your answer
1	SC-3.3	Which is a liquid?	stone / milk / air / pencil
2	SC-3.3	A solid keeps its shape. True or false.	True / False
3	SC-3.3	Ice melting into water is a change from ...	liquid to solid / solid to liquid / gas to liquid / solid to gas
4	SC-3.3	Air in a balloon is a ...	solid / liquid / gas
5	SC-3.3	Water in a freezer becomes ...	steam / ice / air / juice
6	SC-3.3	Name one solid and one liquid in your kitchen.	_____
7	SC-3.2	Plants make their food in their ...	roots / leaves / flowers / seeds
8	SC-3.2	Plants need sunlight to make food. True or false.	True / False
9	SC-3.2	Which is NOT needed by a plant to make food?	sunlight / water / air / sugar from a shop
10	SC-3.2	Leaves are green because of ...	paint / chlorophyll / water / soil

#	Outcome	Question	Your answer
11	SC-3.7	In a fair test we change ...	everything / only one thing / nothing / two things
12	SC-3.7	In our plant test, what did we change?	the soil / the water / the light / the pot
13	SC-3.7	Why must both plants get the same water?	_____

Social Studies

#	Outcome	Question	Your answer
1	SS-3.6	The sun rises in the ...	west / east / north / south
2	SS-3.6	A map key explains the symbols on a map. True or false.	True / False
3	SS-3.6	The capital of Pakistan is ...	Lahore / Karachi / Islamabad / Quetta
4	SS-3.6	Which is a province of Pakistan?	Sindh / Delhi / Dubai / Kabul
5	SS-3.1	Quaid-e-Azam was born in ...	Lahore / Karachi / Sialkot / Quetta
6	SS-3.1	Allama Iqbal is the national poet of Pakistan. True or false.	True / False
7	SS-3.1	Pakistan came into being on ...	23 March 1940 / 14 August 1947 / 25 December 1876 / 9 November 1877
8	SS-3.1	Allama Iqbal was born in ...	Sialkot / Karachi / Peshawar / Multan
9	SS-3.1	Quaid-e-Azam's motto was "Unity, Faith, Discipline". True or false.	True / False
10	SS-3.2	Who makes the laws of the country?	shopkeepers / the government / children / teachers
11	SS-3.2	Throwing litter in the street is a good citizen's habit. True or false.	True / False
12	SS-3.2	Write one duty of a good citizen.	_____